

14.8 Lagrange Multipliers

1 Warm Up

2 Motivation

What if we do not have a bounded region in which we want to find absolute extrema but rather a specific curve or constraint we must live on? Yes we could try to substitute this curve into our surface and turn the problem into a calc-1 problem, but if the curve is complicated, or implicitly defined, this could get real tricky.

3 Definitions & Theorems

1. For a *Lagrange multiplier problem* we are always given a surface and a constraint. More specifically our surface may be $z = f(x, y)$ and our constraint may be $g(x, y) = c$. Notice that the constraint is a single level curve while the surface is an infinite family of level curves.
2. The astute observation is the following: At an absolute extrema, the normal of the constraint level curve will be going in the same direction as the normal of a level curve of our original surface. All of this is to say

$$f_x = \lambda g_x \text{ and } f_y = \lambda g_y$$

We call λ the multiplier constant.

3. Note that this idea scales for higher dimensional surfaces!

4 Examples

Problem 1. *Use Lagrange multipliers to find the extrema of the below surface given the constraint (you do not need to find the extrema's values).*

Maximize: $f(x, y) = x^2 + 4y^2 - 2x + 8y$

Subject to the constraint: $x + 2y = 7$

Problem 2. Use Lagrange multipliers to find the absolute extrema of the function $f(x, y) = 2x^2 + y^2 + x - 3y + 5$ constrained by the level curve $2x - 2y = \frac{11}{2}$.

5 Scratch Work